SHALINI KUMARESAN

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PROFESSIONAL SUMMARY

Fresher with hands-on experience building AI/ML pipelines and Generative AI solutions across four internships and multiple end-to-end projects, including an LLM-based job assistant using LLM APIs, prompt engineering, embeddings, and RAG concepts. Skilled across the ML lifecycle — data preprocessing, model training, evaluation, and deployment-ready inference — using Python, scikit-learn, and TensorFlow, with working knowledge of vector databases, SQL, and agent-based/tool-calling workflows.

SKILLS

Generative AI & LLMs: Prompt Engineering, LLM APIs (OpenAI, Anthropic, Groq/Llama, Azure OpenAI), RAG Architectures, Vector Databases, Embeddings, Agent-Based & Tool-Calling Workflows, Responsible AI Testing & Output Evaluation

Machine Learning & Deep Learning: Regression, Classification, Random Forest, XGBoost, SVM, CNN (TensorFlow), Hyperparameter Tuning, Model Evaluation, Image Preprocessing & Augmentation

Programming & Data: Python (scikit-learn, TensorFlow, NumPy, Pandas), SQL, REST API Integration, Structured & Semi-Structured Data Handling

NLP & Data Engineering: TF-IDF & Cosine Similarity Matching, Data Cleaning, EDA, Feature Engineering, Data Mining, Data Modeling, Web Automation (Playwright)

MLOps & Automation: Git & GitHub, CI/CD Fundamentals, Automation Pipelines, Config & Error Handling, Data Validation

Tools & Platforms: Power BI, Matplotlib, Seaborn, Jupyter Notebook, Google Colab, Anaconda, AWS (IAM Basics)

INTERNSHIP EXPERIENCE

Trainee Software Engineer – Data Analyst — GWC **Apr 2026 – Jun 2026**

- Applied Python, SQL, and Power BI to build AI/ML-adjacent data pipelines, engineer features, and analyze structured datasets.
- Built an automated API-to-BI pipeline integrating REST API ingestion with SQL storage and Power BI visualization, applying config management and error handling.
- Wrote SQL queries to extract, clean, and transform structured data for analysis and downstream AI/ML pipelines.

AI/ML Intern — IBM SkillsBuild **Sep 2024 – Oct 2024**

- Built supervised machine learning models using scikit-learn for predictive analytics on structured datasets.
- Performed data cleaning, feature engineering, and exploratory data analysis (EDA), improving prediction accuracy by 12%.

Data Analysis Intern — Infosys Springboard **Jun 2024 – Jul 2024**

- Conducted data wrangling, statistical analysis, and visualization on structured data using Python.
- Generated analytical reports highlighting key trends to support cross-functional decision-making.

AI/ML Intern — AICTE **Apr 2024 – May 2024**

- Implemented regression and classification models in Python for predictive analysis, applying core ML/model-evaluation concepts.
- Applied hyperparameter tuning and evaluated model performance metrics for accuracy and reliability.

PROJECTS

AI-Powered Job Search & Application Assistant (GenAI / LLM Pipeline) | *Python, Groq API (Llama), Scikit-learn, Pandas, Playwright, REST APIs*

- Built an end-to-end GenAI pipeline ingesting job listings into a unified dataset, letting users search, filter, and chat with an AI assistant for a conversational job search experience.
- Engineered a TF-IDF and cosine-similarity NLP matching engine to score and rank job-resume fit; integrated the Groq LLM API (Llama) with prompt engineering to auto-generate resume-referenced cover letters.
- Built a multi-step, tool-calling workflow — fetch → transform → score → generate → track — orchestrating Playwright-based form-filling with config management and error handling.

Advanced Driver Assistance System (ADAS) using AI & Sensor Fusion | *Python, YOLOv8, OpenCV, Kalman Filtering*

- Developed an ADAS system integrating YOLOv8 object detection, radar/sensor fusion, and Kalman filtering for real-time obstacle detection, collision avoidance, and predictive tracking.

Plant Disease Detection System | *Python, TensorFlow, CNN*

- Developed a deep learning CNN model for plant leaf disease classification with deployment-ready inference, using image preprocessing/augmentation and confusion-matrix evaluation.

Loan Eligibility Prediction Based on Credit History and Other Variables | *Python, Scikit-learn, Anaconda*

- Built a classification model to automate loan-eligibility segmentation using applicant demographics, income, credit history, and loan-amount variables.

Wine Quality Analysis | *Python, Scikit-learn, SVM, Logistic Regression*

- Built classification models (SVM, logistic regression) on the red-wine quality dataset to predict wine quality from chemical attributes, identifying the most influential features.

EDUCATION

B.Tech, Artificial Intelligence and Data Science **2025**
Paavai College of Engineering | CGPA: 8.87